COMMUTATORS WITH THE TRIANGULAR PROJECTION AND HANKEL FORMS ON NEST ALGEBRAS

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Let $\mathscr{B}_p$, $1 \leqslant p < \infty$, denote the von Neumann–Schatten classes and let $\mathscr{B}$ denote the bounded linear operators acting on a separable complex Hilbert space. Let $\mathscr{K}$ denote the compact operators. Associated with every totally ordered family, or nest, of self-adjoint projections in $\mathscr{B}$ there is a nest algebra $\mathscr{A}$ and a transformation $\mathscr{P}$ of lower triangular truncation. It is known that $\mathscr{P}$ possesses boundedness and weak type properties on the classes $\mathscr{B}_p$, $1 < p < \infty$, and on the Schatten–Lorentz classes, respectively, that are analogous to those of the Riesz projection (for functions on the unit circle). See [**12**, **13**, **2**] for example.

We take the parallel with the Riesz projection further. For certain triangular projections of discrete type it is shown that the commutator

$$\mathscr{P}B - B\mathscr{P}$$

determines a compact operator on $\mathscr{B}_2$ if and only if the operator B (acting as a left multiplier) belongs to the C*-algebra

$$(\mathscr{A} + \mathscr{K}) \cap (A + \mathscr{K})^*.$$

This algebra plays the role of the bounded functions on the circle of vanishing mean oscillation (the quasicontinuous functions). For function space contexts see [**33**, **35**, **6**]. The triangular conjugate $\tilde{X}$ of an operator X on l^2 is introduced to provide an alternative description of this C*-algebra. Moreover, a characterisation of $\mathscr{B} + \tilde{\mathscr{B}}$ is given that is analogous to Fefferman's description [**11**] of $L^\infty + \tilde{L}^\infty$ as the functions of bounded mean oscillation. The main idea involved is an 'atomic' decomposition property for the predual of $\mathscr{B} + \tilde{\mathscr{B}}$.

Our approach to commutators involves characterising the bounded bilinear forms $[\,,\,]$ on the Hilbert–Schmidt subspace $\mathscr{A}_2 = \mathscr{B}_2 \cap \mathscr{A}$ that satisfy the identity

$$[A_1 A_2, A_3] = [A_1, A_2 A_3]$$

for all triples in $\mathscr{A}_2$. Such forms are known as Hankel forms. The characterisation is based on a weak factorisation property for the operators in $\mathscr{A}_1$, the triangular trace-class operators, together with the weak star density of the finite-rank operators of $\mathscr{A}$. These facts are related, and the latter, due to Erdos [**8**], is given a new proof. The factorisation property is linked closely to the atomic decomposition mentioned above, to the distance formula of Arveson [**3**, **4**], and to related ideas discussed in [**17**, **27**, **28**, **18**].

An operator X in $\mathscr{B}$ determines a Hankel operator H_X on $\mathscr{A}_2$ such that

$$H_X A = (I - \mathscr{P}) XA$$

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for A in $\mathscr{A}_2$. In the case of a finitely ascending discrete nest of order type $\mathbb{N}$ the compactness of H_X is shown to correspond to the quasitriangularity of the symbol operator X. This connection is a useful one. We deduce that the difference of two truncation operators is compact precisely when their corresponding nests are asymptotic. Also the techniques of Axler, Berg, Jewell and Shields are applicable and we conclude that $\mathscr{A}+\mathscr{K}$ is proximal in this case; that is, every operator possesses a best quasitriangular approximant in the operator norm. More general results on the proximinality of perturbed spaces are obtained in [7].

1. *Weak factorisation and Hankel forms*

Throughout the paper we let $(\mathscr{B}_p, \|\ \|_p)$, $1 \leqslant p < \infty$, denote the von Neumann–Schatten classes of operators that act on a complex separable Hilbert space $\mathscr{H}$. The Banach space of compact operators is denoted by $\mathscr{K}$ and we identify the dual space with $\mathscr{B}_1$ by means of the pairing

$$\langle K, B\rangle = \operatorname{trace}(BK)$$

for B in $\mathscr{B}$ and K in $\mathscr{K}$. The dual space of $\mathscr{B}_1$ is identified with $\mathscr{B}$ in the same manner.

In this section we consider a complete nest $\mathscr{E}$ of self-adjoint projections E on $\mathscr{H}$. Thus $\mathscr{E}$ contains the projections 0 and I, $\mathscr{E}$ is closed in the strong operator topology, and any two projections are comparable with respect to the usual ordering; $F < E$ if and only if $E-F$ is a non-zero positive operator. If $E\in\mathscr{E}$ and $E > 0$ then $E_- = \sup\{F\in\mathscr{E}: F < E\}$. Similarly, if $E\in\mathscr{E}$ and $E < I$, we let $E_+ = \inf\{F\in\mathscr{E}: F > E\}$. If $F > E$ then the projection $F-E$ is called an interval of $\mathscr{E}$. The atoms of $\mathscr{E}$ are the irreducible intervals. The nest algebra $\mathscr{A}$ associated with $\mathscr{E}$ is the set $\{A\in\mathscr{B}: (I-E)AE = 0 \text{ for } E\in\mathscr{E}\}$. This consists of the operators that leave invariant all the subspaces $E\mathscr{H}$, and is often written as $\operatorname{Alg}\mathscr{E}$. We shall write $\mathscr{A}^+$ for the collection of those operators A in $\mathscr{A}$ for which $QAQ = 0$ for all atoms Q. We also let $\mathscr{A}_p = \mathscr{A}\cap\mathscr{B}_p$ and $\mathscr{A}_p^+ = \mathscr{A}_p\cap\mathscr{A}^+$, for $1 \leqslant p < \infty$.

We first obtain a decomposition for operators in $\mathscr{A}_1$ that has proved to be useful [**29, 30**]. We give a quick existential approach to this that is based on the Krein–Millman theorem rather than the constructive methods of [**28**]. Our starting point however is the same fundamental lemma of Lance [**17**].

LEMMA 1.1. *Let A be a trace-class operator that leaves invariant a proper closed subspace, and let E denote the orthogonal projection onto this subspace. Then there exists a trace-class decomposition $A = A_1 + A_2$ such that*

(i) $\|A\|_1 = \|A_1\|_1 + \|A_2\|_1$,

(ii) $A_1 = EA_1$ *and* $A_2E = 0$.

LEMMA 1.2. *The extreme points of the closed unit ball of $\mathscr{A}_1$ are the rank-one operators in the unit sphere. Each such operator has the form $e\otimes f$ where $Ee = 0$ and $E_+f = f$ for some E in $\mathscr{E}$, and where e and f are unit vectors.*

Proof. Let A be an extreme point. First we show that there is a projection E in $\mathscr{E}$ such that $A = E_+A(I-E)$.

Suppose that $E\in\mathscr{E}$, $A \neq EA$ and $AE \neq 0$. Let $A = A_1 + A_2$ be the decomposition of Lemma 1.1 associated with E. Then $A_1E = AE$, and so $A_1 \neq 0$. Also

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$(I-E)A = (I-E)A_2$, and so $A_2 \neq 0$. Thus by Lemma 1.1 (i) A is not an extreme point. This contradiction shows that we have the alternative, $A = EA$ or $AE \neq 0$, for all $E \in \mathscr{E}$. Now let $E = \sup\{F: AF = 0\}$. It follows that $AE = 0$ and $A = GA$ for all $G > E$. Hence $A = E_+ A(I-E)$.

Let $A = \sum \lambda_k R_k$ be a Schmidt decomposition for A, with $\lambda_1, \lambda_2, \ldots$ the singular number sequence of A, and $R_1, R_2, \ldots$ rank-one operators of unit norm. Then

$$A = E_+ A(I-E) = \sum \lambda_k E_+ R_k(I-E).$$

Since $\| A \|_1 = \sum \lambda_k$ it follows that $\lambda_k = \lambda_k \| E_+ R_k(I-E) \|_1$ for all k. In particular $R_k = E_+ R_k(I-E)$ for all $\lambda_k \neq 0$. But this condition on R_k implies membership of $\mathscr{A}$. Since A is an extreme point it follows that $\lambda_2 = \lambda_3 = \ldots = 0$.

To complete the proof observe that a rank-one operator of unit norm is an extreme point in the unit ball of $\mathscr{B}_1$.

LEMMA 1.3. *Let $\varepsilon > 0$ and $A \in \mathscr{A}_1$. Then there exists a sequence $R_1, R_2, \ldots$ of rank-one operators in $\mathscr{A}$ such that*

(i) $A = R_1 + R_2 + \ldots$,
(ii) $\sum \| R_k \|_1 < \| A \|_1 + \varepsilon$.

Proof. In view of the Krein–Millman theorem, Lemma 1.2 and elementary functional analysis, it will be sufficient to show that $\mathscr{A}_1$ is a dual space. Let $\mathscr{S}$ denote the norm-closed linear span of the rank-one operators R such that $R = ER(I-E)$ for some E in $\mathscr{E}$. Then an operator A in $\mathscr{B}_1$ belongs to the annihilator of $\mathscr{S}$ if and only if

$$\operatorname{trace}(X(I-E)AE) = \operatorname{trace}(AEX(I-E)) = 0$$

for all E in $\mathscr{E}$, and all rank-one operators X. It follows that $\mathscr{A}_1$ is the annihilator of $\mathscr{S}$, and thus equal to the dual space of $\mathscr{K}/\mathscr{S}$ through standard duality.

COROLLARY 1.4 [8]. *The finite-rank operators in the operator norm unit ball of a nest algebra are dense in the ultraweak topology.*

Proof. The rank-one operators of $\mathscr{A}$ are described in Lemma 1.2 (this part of the lemma is a well-known and useful fact due to Ringrose [31]). Let $\mathscr{R}^-$ denote the closed linear span of these operators with respect to the operator norm. Then, as in the proof of Lemma 1.3, the annihilator of $\mathscr{R}^-$ in $\mathscr{B}_1$ is $\mathscr{A}_1^+$. Also the operators of $\mathscr{A}_1^+$ admit a decomposition into rank-one operators as in Lemma 1.3. (The proof follows the same pattern.) It is now clear that the annihilator of $\mathscr{A}_1^+$ in $\mathscr{B}$ is equal to the annihilator of the rank-one operators of $\mathscr{A}^+$. But this is the collection of operators A for which

$$\operatorname{trace}(X(I-E)AE) = \operatorname{trace}(EX(I-E)A) = 0$$

for all E in $\mathscr{E}$ and all rank-one operators X, and so coincides with $\mathscr{A}$.

We have shown that $\mathscr{A}$ is the second annihilator of $\mathscr{R}^-$ in the standard duality, and thus is naturally identified with the second dual of $\mathscr{R}^-$. Moreover the weak star topology corresponds to the relative weak star, or ultraweak topology on $\mathscr{A}$. A well-known Banach space principle (sometimes called Goldstine's theorem) now shows that the unit ball of $\mathscr{R}^-$ is weak star, and so ultraweakly, dense in the unit ball of $\mathscr{A}$. The corollary follows.

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REMARKS. The original proof of Corollary 1.4 made use of the representation theory of nests and is quite different in character from the one above. A consequence of the density is the apparently weaker assertion that a nest algebra is local in the sense of [**10**]; that is, the finite-rank operators of $\mathscr{A}$ are ultraweakly dense in the algebra. In fact the Erdos density result may be obtained from localness by using the duality arguments of the proof of Corollary 1.4. However no real simplification arises through this approach since the core of the proof of localness in [**10**] requires Lidskii's theorem that the trace and the spectral trace of a trace-class operator agree. Indeed, it appears to be of more interest to obtain the surprisingly difficult theorem of Lidskii from triangular density properties, as in [**9, 29**]. These ideas seem to be strongly tied to the Hilbert space setting (see [**16, 20**]).

The finite-rank operators of $\mathscr{A}$ are operator norm dense in $\mathscr{A} \cap K$ [**8**]. This simple consequence of Corollary 1.4 is in fact not so deep and may be obtained by direct methods which are valid in wider Banach space contexts (such as the natural nests on $L^p(\mathbb{R}, \mu)$, $1 < p < \infty$) where decomposition theorems for triangular nuclear operators are not at hand.

One of the consequences of localness obtained in [**10**] is that the sum $\mathscr{A} + \mathscr{K}$ is closed. It is amusing to note that this may be obtained directly from Corollary 1.4 and the Banach space arguments of Rudin [**32**] for spaces of type $H^\infty + C$. Approximate identity arguments of this nature also appear in [**19**].

THEOREM 1.5. *Let $\varepsilon > 0$ and $A \in \mathscr{A}_1$. Then there exist rank-one operators $B_1, B_2, \ldots$ and $C_1, C_2, \ldots$ in $\mathscr{A}$ such that*

(i) $A = \sum_k B_k C_k$,

(ii) $\sum_k \| B_k \|_2 \| C_k \|_2 < \| A \|_1 + \varepsilon$.

Proof. Suppose first that $R \in \mathscr{A}_1$ is a non-zero rank-one operator and thus of the form $e \otimes f$ with $Ee = 0$ and $E_+ f = f$ for some $E \in \mathscr{E}$ (Lemma 1.2). If $E < E_+$, let g be a unit vector in the range of $E_+ - E$, so that the operators $B = g \otimes f$ and $C = e \otimes g$ belong to $\mathscr{A}$. Then $R = BC$ and $\| R \|_1 = \| B \|_2 \| C \|_2$. On the other hand, if $E = E_+$, choose $F > E$ so that $\| R - R(I-F) \|_1 < \varepsilon$. Let $R(I-F) = R_1 = e_1 \otimes f$ and choose a unit vector g_1 in the range of $F - E$ so that $B_1 = g_1 \otimes f$ and $C_1 = e_1 \otimes g_1$ belong to $\mathscr{A}$. Then $R_1 = B_1 C_1$ and $\| R_1 \|_1 = \| B_1 \|_2 \| C_1 \|_2$.

In conjunction with Lemma 1.3 the constructions above show that for $\varepsilon > 0$ and A in $\mathscr{A}_1$, there exist rank-one operators $B_1, B_2, \ldots$ and $C_1, C_2, \ldots$ in $\mathscr{A}$ such that

$$\sum_k \| B_k \|_2 \| C_k \|_2 < \| A \|_1 + \varepsilon \quad \text{and} \quad \| A - \sum_k B_k C_k \|_1 < \varepsilon.$$

Iterative use of this principle completes the proof.

A bilinear form $[\,,\,]$ on a complex algebra is called a Hankel form if the identity $[A_1 A_2, A_3] = [A_1, A_2 A_3]$ holds for all triples. A bilinear form $[\,,\,]$ on a normed space is said to be bounded if $|[A_1, A_2]|$ is bounded for all couples A_1, A_2 in the unit ball. Characterisations of bounded Hankel forms on function spaces have been found by Nehari [**22**] for the complex polynomials with the H^2 norm, by Coifman, Rochberg and Weiss [**6**] for complex polynomials in several variables and the Hilbert space norms for the unit sphere and ball, and by Peetre [**23**] for other Bergman space norms. A key step in obtaining these results, as with our next theorem, is the use of weak factorisation ($A = \sum B_k C_k$).

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THEOREM 1.6. *Let* $[\ ,\ ]$ *be a bounded Hankel form on* $\mathscr{A}_2$. *Then there exists a bounded operator* X *such that*

(i) $[A_1, A_2] = \operatorname{trace}(A_2 X A_1)$ *for all* A_1, A_2 *in* $\mathscr{A}_2$,

(ii) $\| X \| = \sup\{|[A_1, A_2]| : \| A_1 \|_2 \leqslant 1, \| A_2 \|_2 \leqslant 1\}$.

Proof. Using Corollary 1.4 fix a bounded sequence R_n of operators in $\mathscr{R}$ that converge to the identity in the ultraweak topology. Let $A = \sum B_k C_k$ be a weak factorisation of an operator A in $\mathscr{A}_1$, as given by Theorem 1.5. Since the series also converges in the Hilbert–Schmidt norm we have

$$\begin{aligned}\sum_k [B_k, C_k] &= \lim_n \sum_k [B_k, C_k R_n] \\ &= \lim_n \sum_k [B_k C_k, R_n] \\ &= \lim_n [\sum_k B_k C_k, R_n] \\ &= \lim_n [A, R_n].\end{aligned}$$

Let us denote this limit by $\Phi(A)$ and thereby define a linear functional on $\mathscr{A}_1$. Thus $\Phi(A) = [A_1, A_2]$ if $A = A_1 A_2$ with A_1, A_2 in $\mathscr{A}_2$. If α denotes the supremum in Theorem 1.6(ii) we have

$$|\Phi(A)| \leqslant \sum_k |[B_k, C_k]| \leqslant \alpha \sum_k \| B_k \|_2 \| C_k \|_2$$

and so it follows from Theorem 1.5 that the norm of Φ is no greater than α. Hence the norm is precisely α. Let X be an operator in $\mathscr{B}$ that implements any norm-preserving extension of Φ to a functional on $\mathscr{B}_1$. With this X the theorem follows.

REMARKS. Let $[\ ,\ ]_X$ denote the Hankel form determined by an operator X in $\mathscr{B}$ and the equation $[A_1, A_2]_X = \operatorname{trace}(A_2 X A_1)$. Then, by weak factorisation, the form is the zero form if and only if X is in the annihilator of $\mathscr{A}_1$, and therefore (as in the proof of Corollary 1.4), if and only if X is in $\mathscr{A}^+$. It now follows from Theorem 1.6 that

$$\sup\{|[A_1, A_2]_X| : \| A_i \|_2 \leqslant 1,\ i = 1, 2\} = \operatorname{dist}(X, \mathscr{A}^+).$$

The quantity on the left is called the norm of the form.

As mentioned earlier, the space $\mathscr{A}_1^+$ also admits a decomposition as in Lemma 1.3, and this leads to the characterisation of the Hankel forms on the product space $\mathscr{A}_2^+ \times \mathscr{A}_2$. In this case the norm of the form implemented by the operator X is $\operatorname{dist}(X, \mathscr{A})$.

The theorem suggests the attractive problem of characterising the bounded Hankel forms on reflexive algebras, both on Hilbert space and general Banach spaces. Because of the close connections with the existence of distance formulas progress will probably depend on new developments in this topic.

2. *Commutators and triangular conjugation*

In this section we specialise to a nest $\mathscr{E}$ that consists of 0, I and an increasing sequence of finite-rank projections $P_1, P_2, \ldots$ that converge in the strong operator topology to the identity. We regard $\mathscr{B}_2$ as a complex Hilbert space with an inner

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product given by $(B_1, B_2) = \operatorname{trace}(B_2^* B_1)$. The triangular projection associated with $\mathscr{E}$ is the orthogonal projection $\mathscr{P}$ of $\mathscr{B}_2$ onto $\mathscr{A}_2$. We write $\mathscr{P}_- = I - \mathscr{P}$ for the complementary projection and $\mathscr{P}_+$ for the orthogonal projection with range $\mathscr{A}_2^+$. A complex linear unitary operator $\mathscr{C}$ is defined on $\mathscr{B}_2$ by the adjoint operation; $\mathscr{C}B = B^*$.

Consider the bounded Hankel form $[A_1, A_2] = \operatorname{trace}(A_2 X A_1)$ that is induced by a bounded operator X. If $A_1 \in \mathscr{A}_2^+$ and $A_2 \in \mathscr{A}_2$ then

$$\begin{aligned}[A_2, A_1]_X &= \operatorname{trace}(X A_2 A_1) \\ &= (A_1, (X A_2)^*) \\ &= (\mathscr{P}_+ A_1, \mathscr{C}(X A_2)) \\ &= (A_1, \mathscr{P}_+ \mathscr{C}(X A_2)) \\ &= (A_1, \mathscr{C} H_X A_2),\end{aligned}$$

where H_X is the Hankel operator $(I-\mathscr{P})X\mathscr{P}$. The Hankel operator belongs to $\mathscr{B}(\mathscr{B}_2)$, and we can see from the above that its operator norm coincides with the norm of the Hankel form on $\mathscr{A}_2 \times \mathscr{A}_2^+$. Thus by our remarks in Section 1 we have

$$\| H_X \| = \operatorname{dist}(X, \mathscr{A}).$$

THEOREM 2.1. *Let X be a bounded operator. Then*

(i) *the Hankel operator H_X is a compact operator if and only if X belongs to the quasitriangular algebra $\mathscr{A} + \mathscr{K}$. Moreover*

$$\operatorname{dist}(H_X, \mathscr{K}(\mathscr{B}_2)) = \operatorname{dist}(X, \mathscr{A} + K);$$

(ii) *the commutator $X\mathscr{P} - \mathscr{P}X$ determines a compact operator on $\mathscr{B}_2$ if and only if X belongs to the* C*-*algebra*

$$(\mathscr{A} + \mathscr{K}) \cap (\mathscr{A} + \mathscr{K})^*.$$

Proof. Note that $H_X = 0$ if $X \in \mathscr{A}$. Also, if $X = P_n X P_n$ for some n, then H_X has finite rank. Indeed, if $A \in \mathscr{A}_2$, then $XA(I-P_n) \in \mathscr{A}_2$, and so the range of H_X is contained in $(I-\mathscr{P})X\mathscr{A}_2 P_n = (I-\mathscr{P})XP_n \mathscr{A}_2 P_n$, which is finite dimensional. Since $\| H_X \| \leqslant \| X \|$ it follows that H_X is a compact operator when $X \in \mathscr{K}$, and so too when $X \in \mathscr{A} + \mathscr{K}$.

Suppose now that H_X is a compact operator. For $n = 1, 2, \ldots$ let $S_n = n^{-1}P_n + (I - P_n)$ so that S_n is a bounded sequence of invertible operators that converge to zero in the strong operator topology. The operator of left multiplication by S_n on $\mathscr{B}_2$ also converges to zero in the strong operator topology (of $\mathscr{B}(\mathscr{B}_2)$) and so, using the compactness of H_X, we see that $H_{XS_n} = H_X S_n$ converges to zero in operator norm. By the identity preceding Theorem 2.1 there exist operators $A_n \in \mathscr{A}$ such that the norm of $XS_n + A_n$ tends to zero as n tends to infinity. Let $\pi(T)$ denote the coset of T in the Calkin algebra $\mathscr{B}/\mathscr{K}$. Then

$$\| \pi(X + A_n S_n^{-1}) \| \leqslant \| \pi(XS_n + A_n) \| \, \| \pi(S_n^{-1}) \|$$

and $\| \pi(S_n^{-1}) \| \leqslant 1$. Since $A_n S_n^{-1} \in \mathscr{A} + \mathscr{K}$, it follows that $X \in \mathscr{A} + \mathscr{K}$.

Now let $K \in \mathscr{K}(\mathscr{B}_2)$, so that the operator KS_n converges to zero, where once again S_n is regarded as a left multiplier. For large enough n, we have

$$\begin{aligned}\| H_X + K \| &\geqslant \| H_{XS_n} + KS_n \| \geqslant \| H_{XS_n} \| - \varepsilon \\ &= \operatorname{dist}(XS_n, \mathscr{A}) - \varepsilon \geqslant \operatorname{dist}(X, \mathscr{A} + \mathscr{K}) - \varepsilon.\end{aligned}$$

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On the other hand, by our opening comments, if $Y \in \mathscr{A}+\mathscr{K}$, we have

$$\operatorname{dist}(H_X, \mathscr{K}(\mathscr{B}_2)) = \operatorname{dist}(H_{X+Y}, \mathscr{K}(\mathscr{B}_2) \leqslant \| X+Y \|,$$

and so the proof of (i) is complete.

(ii) Note that

$$X\mathscr{P}-\mathscr{P}X = (X\mathscr{P}-\mathscr{P}X\mathscr{P})-(\mathscr{P}X-\mathscr{P}X\mathscr{P}) = H_X-(H_{X^*})^*,$$

so that, by (i), the condition on X is sufficient for compactness. On the other hand, if the commutator is compact then so too are the operators $\mathscr{P}X\mathscr{P}-\mathscr{P}X$ and $X\mathscr{P}-\mathscr{P}X\mathscr{P}$. Thus, by (i) again, the condition is necessary.

It has been shown by Plastiras [25] that another finitely ascending nest, $Q_1, Q_2, \ldots$ say, determines the quasitriangular algebra $\mathscr{A}+\mathscr{K}$ if and only if $\{P_n\}$ and $\{Q_n\}$ are asymptotic. This means that P_n-Q_{n+k} converges to zero in norm, as n tends to infinity, for some fixed integer k. Ken Davidson noticed the following consequence.

COROLLARY 2.2. *Let $\mathscr{P}$ and $\mathscr{Q}$ be the projections of triangular truncation determined by the finitely ascending nests $P_1, P_2, \ldots$ and $Q_1, Q_2, \ldots$ respectively. Then $\mathscr{P}-\mathscr{Q}$ is a compact operator if and only if $\| P_n-Q_{n+k} \| \to 0$, as $n \to \infty$, for some integer k.*

Proof. Let $E_n = P_n-P_{n-1}$ and $F_n = Q_n-Q_{n-1}$. For an operator X we have

$$\begin{aligned}(\mathscr{P}-\mathscr{Q})X &= \sum_{k=1} P_k X E_{k+1} - \sum_{k=1} Q_k X F_{k+1} \\ &= \sum_{k=1} (P_k-Q_k) X E_{k+1} + \sum_{k=1} Q_k X (E_{k+1}-F_{k+1}) \\ &= \sum_{k=1} (P_k-Q_k) X E_{k+1} + \sum_{k=1} Q_k X (P_{k+1}-Q_{k+1}) - \sum_{k=1} Q_k X (P_k-Q_k).\end{aligned}$$

Since the projections E_n and Q_n have finite rank it follows that $\mathscr{P}-\mathscr{Q}$ is a compact operator if the nests are asymptotic.

On the other hand, if $\mathscr{P}$ is a compact perturbation of $\mathscr{Q}$ then the Hankel operator $(I-\mathscr{P})X\mathscr{P}$ is compact if and only if the Hankel operator $(I-\mathscr{Q})X\mathscr{Q}$ is compact. By Theorem 2.1 the quasitriangular algebras for $\mathscr{P}$ and for $\mathscr{Q}$ coincide, and so, by our earlier comments, $\{P_n\}$ and $\{Q_n\}$ are asymptotic.

REMARKS. (1) If X is a bounded operator and $X_n = P_n X P_n$, then the bounded sequence H_{X_n} of finite-rank Hankel operators converges to H_X in the strong topology. Because of this the results of [5] may be applied to show that there exists a compact Hankel operator H_Y such that

$$\| H_X-H_Y \| = \operatorname{dist}(H_X, \mathscr{K}(\mathscr{B}_2)).$$

In conjunction with the equality $\| H_X \| = \operatorname{dist}(X, \mathscr{A})$ this leads to the fact that $\mathscr{A}+\mathscr{K}$ is proximinal; that is, every operator possesses a best approximation in the operator norm from the quasitriangular algebra $\mathscr{A}+\mathscr{K}$. The proof follows that of [5] concerning the proximinality of $H^\infty+C$ in L^∞. However, there are rather more direct methods available, including the M-ideal techniques of Alfsen and Effros [1], and these are discussed in [7]. These methods cover the context of quasitriangular algebras with respect to an arbitrary nest, as well as certain Banach space contexts. The M-ideal approach was exploited by Leucking [21] to obtain the proximinality of $H^\infty+C$.

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(2) The first part of the theorem yields another proof that the operators X that are quasitriangular with respect to $P_1, P_2, \ldots$ ($(I-P_n)XP_n \to 0$ as $n \to \infty$) are precisely the operators in the quasitriangular algebra $\mathscr{A}+\mathscr{K}$. Indeed, let Q_n denote the orthogonal projection of $\mathscr{B}_2$ onto the subspace $\mathscr{B}_2(P_n-P_{n-1})$. Then $H_X Q_n = Q_n H_{(I-P_n)XP_n} Q_n$. Consequently

$$H_X \oplus 0 = \bigoplus_{n=1}^{\infty} H_{(I-P_n)XP_n}.$$

Since the symbol operators for the summands are of finite rank, the summands themselves are compact operators, and so, by the quasitriangularity of X, it follows that H_X is compact.

(3) The equalities at the beginning of this section show that H_X is closely related to the Hankel operator S_X on $\mathscr{A}_2$ that is defined by

$$S_X : A \longrightarrow \mathscr{P}(XA)^*.$$

For example, S_X is compact if and only if the symbol operator X belongs to $\mathscr{A}^+ + \mathscr{K}$. This assertion does not hold for more general nests, such as the Volterra nest for $L^2[0, 1]$. In the case of the standard multiplicity-one nest of order type $\mathbb{N}$ or $\mathbb{Z}$ the following characterisations can be made:

(i) S_X is of finite rank if and only if the representing matrix of $\mathscr{P}X^*$ is finitely non-zero;

(ii) $S_X \geqslant 0$ if and only if X is a positive diagonal operator modulo $\mathscr{A}^+$;

(iii) S_X is a Hilbert–Schmidt operator if and only if

$$\sum_{j \geqslant k} (j-k+1)\,|x_{jk}|^2$$

is finite, where $X = (x_{jk})$.

More generally, it is possible to use the decomposition for H_X above to characterise when H_X and S_X belong to a given von Neumann–Schatten class. The corresponding characterisations for the classical Hankel operators are due to Peller [**24**].

$\mathscr{P}$-triangular conjugates

There is a strong formal similarity between Theorem 2.1 and certain function space settings involving classical Hankel operators, commutators, and the space $(H^\infty + C) \cap (\overline{H^\infty + C})$. See [**14**, **33**, **26**]. This mirroring can be taken further with the notion of the triangular conjugate of an operator.

Let $\mathscr{E}$ be a nest of multiplicity one and order type $\mathbb{N}$ or $\mathbb{Z}$, and let $\mathscr{F}$ denote the associated linear space of matrices with only a finite number of non-zero entries. The dual space of $\mathscr{F}$ of (conjugate) linear functionals is identified with the space $\mathscr{M}$ of all matrices under the pairing $(M, F) = \text{trace}\,(F^* M)$, for F in $\mathscr{F}$ and M in $\mathscr{M}$. The $\mathscr{P}$-triangular conjugate of a matrix M is defined to be the matrix

$$\tilde{M} = (-i\mathscr{P}_+ + i\mathscr{P}_-)\,M.$$

Note that for every matrix M the matrix $M + i\tilde{M}$ is upper triangular, and if M has zero diagonal, then $M + i\tilde{M} = 2\mathscr{P}_+ M$. If we let $\mathscr{D}$ denote the linear space of diagonal matrices, it follows easily from these facts that

$$(\mathscr{A}+\mathscr{K}) \cap (\mathscr{A}+\mathscr{K})^* = \mathscr{D} \cap \mathscr{B} + (\mathscr{K} + \tilde{\mathscr{K}}) \cap \mathscr{B}.$$

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This provides an alternative description of the C*-algebra of Theorem 2.1. In fact we can give an analogue of Fefferman's characterisation [11] of $L^\infty + \tilde{L}^\infty$ as the functions of bounded mean oscillation. The identification of $\mathscr{B}+\tilde{\mathscr{B}}$ with the dual space of $\mathscr{F}_1$ follows elementary duality principles as in the function space setting. However, the realisation of $B+\tilde{B}$ as a certain space of matrices for which the 'oscillation' $\| ME-EM \|$ is bounded, for $E \in \mathscr{E}$, lies as deep as the decomposition result of Lemma 1.3. We see these facts in the next theorem.

For a matrix M in $\mathscr{B}+\tilde{\mathscr{B}}$ define the norm

$$\| M \|_* = \inf\{\max(\| F \|, \| G \|) : M = F+\tilde{G},\ F, G \in \mathscr{B}\},$$

so that $(\mathscr{B}+\tilde{\mathscr{B}}, \| \|_*)$ is a Banach space. Let $\mathscr{F}_1$ denote the completion of $\mathscr{F}$ with respect to the norm

$$\| F \|_{\mathscr{F}_1} = \| F \|_1 + \| \tilde{F} \|_1.$$

THEOREM 2.3. *The following conditions are equivalent for a matrix M:*

(i) *M belongs to $\mathscr{B}+\tilde{\mathscr{B}}$;*

(ii) *M determines a continuous functional on the Banach space $\mathscr{F}_1$;*

(iii) *the diagonal of M is bounded and the set of commutators $ME-EM$, for E in $\mathscr{E}$, is uniformly bounded in operator norm.*

Also the Banach space $\mathscr{B}+\tilde{\mathscr{B}}$ is isometrically isomorphic to the dual of $\mathscr{F}_1$.

Proof. (i) $\Rightarrow$ (ii) If $M = B_1 + \tilde{B}_2$ with B_1, B_2 in $\mathscr{B}$, and if F is a matrix of $\mathscr{F}$, then

$$|(M, F)| = |(B_1+\tilde{B}_2, F)| = |(B_1, F)-(B_2, \tilde{F})| \leqslant \| B_1 \| \| F \|_1 + \| B_2 \| \| \tilde{F} \|_1.$$

Thus $|(M, F)| \leqslant \| M \|_* \| F \|_{\mathscr{F}_1}$.

(ii) $\Rightarrow$ (i) Let Λ be the linear map between the Banach spaces $\mathscr{B} \oplus \mathscr{B}$ and $\mathscr{B}+\tilde{\mathscr{B}}$ such that $(B_1 \oplus B_2) \to B_1 + \tilde{B}_2$. Then the induced mapping on the quotient space $(\mathscr{B} \oplus \mathscr{B})/\ker \Lambda$ is an isometrical isomorphism. It follows that the predual of $\mathscr{B}+\tilde{\mathscr{B}}$ is naturally identifiable with the annihilator of $\ker \Lambda$ in the predual $\mathscr{B}_1 \oplus \mathscr{B}_1$. However the operator $B_1 \oplus B_2$ belongs to this kernel if and only if $\tilde{B}_2 = -B_1$, which is the condition $B_2 = \tilde{B}_1$. Thus the annihilator consists of operators $C_1 \oplus C_2$ such that

$$\operatorname{trace}((C_1+\tilde{C}_2)B) = \operatorname{trace}(C_1 B + C_2 \tilde{B}) = 0$$

for all B in $\mathscr{B}$ with zero diagonal and with $\tilde{B}$ in $\mathscr{B}$. Hence the annihilator is identified with the subspace of elements of the form $-\tilde{C}_2 \oplus C_2$. Clearly this subspace is isomorphic to $\mathscr{F}_1$.

(ii) $\Rightarrow$ (iii) Let the matrix M determine a $\| \|_{\mathscr{F}_1}$-continuous functional of norm $\alpha > 0$. Let $R = e \otimes f$, where e and f are unit vectors such that $Ee = 0$ and $E_+ f = f$, for some E in $\mathscr{E}$. Since R belongs to $\mathscr{A}$, it follows that $\| \tilde{R} \|_1 \leqslant 2$ and $\| R \|_{\mathscr{F}_1} \leqslant 3$. We have

$$(M, R) = \operatorname{trace}(R^* M) = (Me, f) = (E_+ M(I-E)e, f)$$

and so $\| E_+ M(I-E) \| \leqslant 3\alpha$. Since $\| F \|_{\mathscr{F}_1} = \| F^* \|_{\mathscr{F}_1}$ it follows that $\| E_+ M^*(I-E) \| \leqslant 3\alpha$. The boundedness of these norms is equivalent to property (iii).

(iii) $\Rightarrow$ (ii) By the assertions above, if (iii) holds for a matrix M then for some $\beta > 0$ we have $|(M, R)| \leqslant \beta$ and $|(M, R^*)| \leqslant \beta$ for all R in $\mathscr{A}$ of rank one and unit operator norm. Let F be a self-adjoint operator in the unit ball of $\mathscr{F}_1$. Let $\sum R_k$ denote

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a decomposition of $F+i\tilde{F}$, as provided by Lemma 1.3, with $\sum \| R_k \| \leqslant 2$. Then $F = \sum A_k$, where $A_k = \frac{1}{2}(R_k + R_k^*)$, and so

$$|(M, F)| \leqslant \tfrac{1}{2}\sum (|(M, R_k)| + |(M, R_k^*)|) \leqslant \beta;$$

(ii) now follows.

REMARKS. The last part of the proof shows that elements of the Banach space $\mathscr{F}_1$ admit an atomic decomposition (in the sense of harmonic analysis) into sums of operators of bounded rank.

A constructive approach to the $(L^\infty + \tilde{L}^\infty)$-representation of a function of bounded mean oscillation has been given by Jones [**15**]. It would be interesting to discover an operator-theoretic variant of this process and thereby give a direct proof of the implication (iii) $\Rightarrow$ (i), and possibly provide insight into how the Arveson distance formula is attained.

Like BMO, the space $\mathscr{B}+\tilde{\mathscr{B}}$ has the following monotonicity property. If $0 \leqslant |y_{ij}| \leqslant x_{ij}$ and the matrix $X = (x_{ij})$ belongs to $\mathscr{B}+\tilde{\mathscr{B}}$, then so does the matrix $Y = (y_{ij})$. This result follows from weak factorisation and is dual to the Hardy inequality of Shields [**34**] (see also [**30**]).

Note added in proof. T. G. Feeman has also obtained the proximality of quasi-triangular algebras associated with discrete nests (in a paper to appear in *Trans. Amer. Math. Soc.*). Theorem 1.6(i) has been generalised to semifinite factors in the author's preprint 'Factorisation in analytic operator algebras'.

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